Noah Zhang

(+1) 404-703-7455 | zhangnuoya969@gmail.com | Atlanta, GA
noahz969.github.io | LinkedIn | Google Scholar

Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Physics (Advisor: Surabhi Sachdev) — *Aug 2023 – Present (Exp. 2028)*

Georgia Institute of Technology — Atlanta, GA
M.S. in Computational Science and Engineering (CSE) — *Aug 2023 – May 2025*

Nanyang Technological University (NTU) — Singapore
B.Sc. in Physics (First Class Honors) — *Aug 2019 – May 2023*

- **GPA: 4.62/5.00** (Major GPA: 5.00/5.00). Awarded Full Ministry of Education Scholarship.

Technical Skills & Honors

Statistics: Bayesian Inference, MCMC, Nested Sampling, MLE, Hypothesis Testing, Bootstrap, Permutation Tests, FDR
Machine Learning: Regularized Regression, Time-Series Forecasting, Walk-Forward & Nested CV, PCA, Clustering, Kalman Filtering
Programming & Data: Python (NumPy, SciPy, pandas, statsmodels), C++, SQL; TB-scale streaming data, HPC/HTCondor, Git, Linux
Honors: PLANCKS Physics Competition (Singapore) Silver Medal, 2022 | Odyssey Research Symposium Best Presentation, 2020

Research Experience

LIGO Scientific Collaboration (Georgia Tech) — Aug 2023 – Present
Graduate Research Assistant – Gravitational-Wave Detection & Inference — *Atlanta, GA*

- **Targeted Follow-up Search:** Co-developed a targeted search for the GstLAL pipeline that folds external electromagnetic triggers (arrival time, sky location) into a likelihood-ratio detection statistic as **Bayesian priors**, reducing the false-alarm rate of associated candidates by **orders of magnitude** at unchanged latency.
- **Deployment:** Used in the **O4a** observing run and being deployed for the next run (IR1); now extending it to multi-pixel sky-localization priors. Second author of the method paper (under review, ApJ).
- **Real-Time Parameter Estimation:** Improved minutes-latency source parameter estimation from GstLAL low-latency SNR data by combining **MCMC**, **nested sampling** and prior updating, raising P-P plot calibration consistency in validation by **33%**.
- **IMBH Search (Lead, O4b/O4c):** Lead the offline intermediate-mass black hole search; improved the background noise model using **time-slid triggers**, making the likelihood-ratio ranking statistic more reliable. Contributed to the **GWTC-5** and **GWTC-6** catalogs.
- **Pipeline Engineering:** Maintain the GstLAL hybrid **C++/Python** pipeline, processing terabytes of streaming data with sub-second latency for real-time trigger generation.

Nanyang Technological University — May 2020 – Aug 2022
Undergraduate Researcher – Stochastic Modeling & Simulation — *Singapore*

- Modeled foraging dynamics with correlated random walks and time-delay ODEs; combined MCMC with a customized random-walk sampler, cutting simulation runtime from **days to hours (∼50x)**.

Publications & Presentations

L. Tsukada, **N. Zhang**, et al., *Fast targeted gravitational-wave followup search for compact binary mergers using GstLAL pipeline*, arXiv:2606.22266 (under review, ApJ).
N. Zhang and Yong, *Dynamics, Statistics, and Task Allocation of Foraging Ants*, Phys. Rev. E **108**, 054306 (2023).
LIGO–Virgo–KAGRA Collaboration, *GWTC-5* and *GWTC-6* catalog papers (contributor, IMBH search).
Talk: *Toward fast targeted gravitational-wave followup search for compact binary coalescences using GstLAL inspiral pipeline*, APS Meeting, Mar 18, 2026.

Selected Data Science Projects

Time-Series Forecasting with an Empirical Null | *Python, Ridge Regression* — Sep 2026

- Built a walk-forward forecasting pipeline for daily equity-index returns with leakage-safe splits and train-only standardization; out-of-sample information coefficient **0.088**, **3.2**σ above a block-permutation null ($p < 0.005$) that preserves autocorrelation.

Reproducible Research Pipeline on CRSP Stock Data | *Python, pandas, statsmodels, pytest* | GitHub — Apr 2026 – May 2026

- **Pre-registered** analysis; rebuilt a point-in-time S&P 500 universe by entity-matching external data to CRSP, with look-ahead tests. Replicated signal correlates **0.90** in-sample and **0.83** on a single-shot holdout with the published benchmark.

WorldQuant Int'l Quant Championship 2026 – Stage 2 | *Feature Research* — Mar 2026 – May 2026

- Reached Stage 2 among **156,000+** registrants; 11 predictive signals on 3,000 U.S. stocks (out-of-sample Sharpe **1.3–2.0**); ablation exposed a composite collapsing **1.59** $\rightarrow$ $-$**0.35** without one feature.